
Berkovich Space

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1 Spectrum

Let $\mathbf{k}$ be a spherically complete non-archimedean field which is algebraically closed and $A = \mathbf{k}[T]$. We want to consider the “analytic structure” on $\mathbf{mSpec} A$. However, unlike the complex case, the set $\mathbf{mSpec} A$ is totally disconnected with respect to the topology induced by the absolute value on $\mathbf{k}$ (?). To overcome this difficulty, Berkovich uses multiplicative semi-norms to “fill in the gaps” between the points in $\mathbf{mSpec} A$, leading to the notion of the spectrum of a Banach ring.

1.1 Definition and examples

Definition 1.1. Let R be a Banach ring. The *Berkovich spectrum* $\mathcal{M}(R)$ of R is defined as the set of all multiplicative semi-norms on R that are bounded with respect to the given norm on R . For every point $x \in \mathcal{M}(R)$, we denote the corresponding multiplicative semi-norm by $|\cdot|_x$.

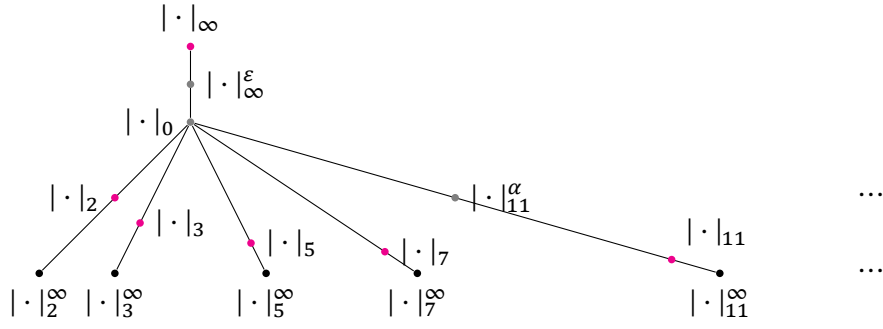
We equip $\mathcal{M}(R)$ with the weakest topology such that for each $f \in R$, the evaluation map $\mathcal{M}(R) \rightarrow \mathbb{R}_{\geq 0}$, defined by $x \mapsto |f|_x =: f(x)$, is continuous.

Example 1.2. Let $(\mathbf{k}, |\cdot|)$ be a complete valuation field. The Berkovich spectrum $\mathcal{M}(\mathbf{k})$ consists of a single point corresponding to the given absolute value $|\cdot|$ on $\mathbf{k}$.

Example 1.3. Consider the Banach ring $(\mathbb{Z}, \|\cdot\|)$ with $\|\cdot\| = |\cdot|_{\infty}$ is the usual absolute value norm on $\mathbb{Z}$. Let $|\cdot|_p$ denote the p -adic norm for each prime number p , i.e., $|n|_p = p^{-v_p(n)}$ for each $n \in \mathbb{Z}$, where $v_p(n)$ is the p -adic valuation of n . The Berkovich spectrum

$$\mathcal{M}(\mathbb{Z}) = \{|\cdot|_{\infty}^{\varepsilon} : \varepsilon \in (0, 1]\} \cup \{|\cdot|_p^{\alpha} : p \text{ is prime}, \alpha \in (0, \infty]\} \cup \{|\cdot|_0\},$$

where $|a|_p^{\infty} := \lim_{\alpha \rightarrow \infty} |a|_p^{\alpha}$ for each $a \in \mathbb{Z}$ and $|\cdot|_0$ is the trivial norm on $\mathbb{Z}$.



Yang: To be continued.

Theorem 1.4. Let R be a Banach ring. The spectrum $\mathcal{M}(R)$ is nonempty.

Proof. Let $\mathfrak{m}$ be a maximal ideal of R . Note that the pullback of residue norm on the residue field $R/\mathfrak{m}$ is bounded with respect to the given norm on R . Replacing R by the completion of $R/\mathfrak{m}$, we may assume that R is a complete field. Consider the set

$$\Sigma = \{\text{norm on } R \text{ bounded by the given norm } \|\cdot\|\}$$

with the partial order defined by boundedness. Since for a descending chain in Σ , the infimum is a norm, by Zorn's lemma, there exists a minimal element $|\cdot| \in \Sigma$.

We claim that $|\cdot|$ is multiplicative. Since the spectral radius $\rho(f) = \lim_{n \rightarrow \infty} |f^n|^{1/n}$ associated to $|\cdot|$ is power-multiplicative and bounded by $|\cdot|$, by minimality of $|\cdot|$, we have $\rho(f) = |f|$ for each $f \in R$. Thus $|\cdot|$ is power-multiplicative. If $|\cdot|$ is not multiplicative, then there exist $a, b \in R \setminus \{0\}$ such that $|ab| < |a||b|$. Then $|b| \leq |a^{-1}||ab| < |a^{-1}||a||b|$, which implies that $|a||a^{-1}| > 1$. Set $r = |a|^{-1} < |a^{-1}|$ and consider $R\langle T/r \rangle$. Since $r \cdot |a| = 1$, we have that

$$\left| \sum_{n=0}^{\infty} a^n T^n \right| = \sum_{n=0}^{\infty} |a^n| r^n = \sum_{n=0}^{\infty} |a|^n r^n = \sum_{n=0}^{\infty} 1 = \infty.$$

The power series is not convergent in $R\langle T/r \rangle$ and hence $1 - aT$ is not invertible in $R\langle T/r \rangle$. Let $\mathfrak{n}$ be a maximal ideal of $R\langle T/r \rangle$ containing $1 - aT$. Consider $R \rightarrow R\langle T/r \rangle \rightarrow R\langle T/r \rangle/\mathfrak{n}$. Since R is a field, the composition is injective. The residue norm on $R\langle T/r \rangle/\mathfrak{n}$ induces a norm $|\cdot|'$ on R bounded by $|\cdot|$. Note that $|a^{-1}|' \leq |T| = r = |a|^{-1} < |a^{-1}|$, contradicting the minimality of $|\cdot|$. $\square$

Definition 1.5. Let $\varphi : R \rightarrow S$ be a bounded ring homomorphism of Banach rings. The *pullback* map $\mathcal{M}(\varphi) : \mathcal{M}(S) \rightarrow \mathcal{M}(R)$ is defined by $\mathcal{M}(\varphi)(x) = x \circ \varphi : f \mapsto |\varphi(f)|_x$ for each $x \in \mathcal{M}(S)$.

Note that $\mathcal{M}(\varphi)(f^{-1}(V)) = \varphi(f)^{-1}(V)$ for each $f \in R$ and open subset $V \subset \mathbb{R}_{\geq 0}$. Hence the pullback map is continuous.

Notation 1.6. Let R be a Banach ring and $x \in \mathcal{M}(R)$. We denote by $|\cdot|_x$ the multiplicative semi-norm on R corresponding to the point x . Its kernel $\{f \in R : |f|_x = 0\}$ is a closed prime ideal of R , denoted by $\wp_x$.

Definition 1.7. Let R be a Banach ring. For each $x \in \mathcal{M}(R)$, the *completed residue field* at the point x is defined as the completion of the residue field $\kappa(x) = \text{Frac}(R/\wp_x)$ with respect to the multiplicative norm induced by the semi-norm $|\cdot|_x$, denoted by $\mathcal{H}(x)$.

Example 1.8. Consider the Banach ring $(\mathbb{Z}, |\cdot|_\infty)$ as in [Example 1.3](#). We have

- $x = |\cdot|_\infty^\varepsilon$ for some $\varepsilon \in (0, 1]$: $\wp_x = (0)$ and $\mathcal{H}(x) \cong \mathbb{R}$ with the absolute value norm raised to the power ε ;
- $x = |\cdot|_0$: $\wp_x = (0)$ and $\mathcal{H}(x) \cong \mathbb{Q}$ with the trivial norm;
- $x = |\cdot|_p^\alpha$ for some prime number p and $\alpha \in (0, \infty)$: $\wp_x = (0)$ and $\mathcal{H}(x) \cong \mathbb{Q}_p$ with the p -adic norm raised to the power α ;
- $x = |\cdot|_p^\infty$ for some prime number p : $\wp_x = (p)$ and $\mathcal{H}(x) \cong \mathbb{F}_p$ with the trivial norm.

Definition 1.9. Let R be a Banach ring. The *Gel'fand transform* of R is the bounded ring homomorphism

$$\Gamma : R \rightarrow \prod_{x \in \mathcal{M}(R)} \mathcal{H}(x), \quad f \mapsto (f(x))_{x \in \mathcal{M}(R)},$$

where the norm on the product $\prod_{x \in \mathcal{M}(R)} \mathcal{H}(x)$ is given by the supremum norm.

Example 1.10. Let K be a number field equipped with [Yang: To be continued](#).

Proposition 1.11. The Gel'fand transform $\Gamma : R \rightarrow \prod_{x \in \mathcal{M}(R)} \mathcal{H}(x)$ of a Banach ring R factors through the uniformization R^u of R , and the induced map $R^u \rightarrow \prod_{x \in \mathcal{M}(R)} \mathcal{H}(x)$ is an isometric embedding. [Yang: To be checked](#).

Proof. [Yang: To be added](#). □

Lemma 1.12. Let $\{K_i\}_{i \in I}$ be a family of completed fields. Consider the Banach ring $R = \prod_{i \in I} K_i$ equipped with the product norm. The spectrum $\mathcal{M}(R)$ is homeomorphic to the Stone-Čech compactification of the discrete space I .

Proof. [Yang: To be added](#). □

Remark 1.13. Here we list some facts about ultrafilters and the Stone-Čech compactification that will be used in the proof of [Lemma 1.12](#).

Let I be a set. A *filter* on I is a nonempty collection $\mathcal{F} \subset \mathcal{P}(I)$ of subsets of I such that:

- if $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$;
- if $A \in \mathcal{F}$ and $A \subset B \subset I$, then $B \in \mathcal{F}$.

For example, let $I = \mathbb{N}$ and

$$\mathcal{F} = \{A \subset \mathbb{N} : \#(\mathbb{N} \setminus A) < \infty\}.$$

Then $\mathcal{F}$ is a filter on $\mathbb{N}$, called the *Fréchet filter*. Another example is the filter called the *principal filter* generated by a point $i_0 \in I$:

$$\mathcal{F}_{i_0} = \{A \subset I : i_0 \in A\}.$$

Filters are partially ordered by inclusion. A filter $\mathcal{U}$ is called an *ultrafilter* if it is maximal with respect to inclusion. A filter is an ultrafilter if and only if for every subset $A \subset I$, either $A \in \mathcal{U}$ or $I \setminus A \in \mathcal{U}$.

One can define the notion of convergence along ultrafilters in a topological space. Let X be a topological space, $(x_i)_{i \in I}$ a collection of points in X indexed by a set I , and $\mathcal{U}$ an ultrafilter on I . We say that $(x_i)_{i \in I}$ *converges* to a point $x \in X$ along the ultrafilter $\mathcal{U}$ if for every open neighborhood U of x , the set $\{i \in I : x_i \in U\}$ belongs to $\mathcal{U}$. It is denoted by $\lim_{\mathcal{U}} x_i = x$. When $I = \mathbb{N}$ and $\mathcal{U}$ is the Fréchet filter, this notion of convergence coincides with the usual notion of convergence of sequences.

The Stone-Čech compactification of a discrete space is the largest compact Hausdorff space in which the original space can be densely embedded. Yang: To be checked.

Theorem 1.14. Let R be a Banach ring. The spectrum $\mathcal{M}(R)$ is a compact Hausdorff space.

Proof. Yang: To be added. □

Corollary 1.15. Let R be a Banach ring and R^u its uniformization. The canonical map $\mathcal{M}(R^u) \rightarrow \mathcal{M}(R)$ induced by the bounded homomorphism $R \rightarrow R^u$ is a homeomorphism.

Proposition 1.16. Let $\mathbf{l}/\mathbf{k}$ be a finite normal extension of complete valuation fields and A a Banach $\mathbf{k}$ -algebra. Then we have:

- (a) if $\mathbf{l}/\mathbf{k}$ is Galois, then $\mathcal{M}(A \otimes_{\mathbf{k}} \mathbf{l})$ admits a natural action of the Galois group $\text{Gal}(\mathbf{l}/\mathbf{k})$ and the quotient space is homeomorphic to $\mathcal{M}(A)$;
- (b) if $\mathbf{l}/\mathbf{k}$ is purely inseparable, then the canonical map $\mathcal{M}(A \otimes_{\mathbf{k}} \mathbf{l}) \rightarrow \mathcal{M}(A)$ is a homeomorphism.

Proof. Yang: To be added. □

Corollary 1.17. Let $\mathbf{k}$ be a complete valuation field, $\mathbf{l}$ its completion of an algebraic closure, and A a Banach $\mathbf{k}$ -algebra. Then we have a natural homeomorphism $\mathcal{M}(A \hat{\otimes}_{\mathbf{k}} \mathbf{l}) / \text{Gal}(\mathbf{l}/\mathbf{k}) \cong \mathcal{M}(A)$.

Proof. Yang: To be added. □

1.2 Reduction map and kernel map

Proposition 1.18. Let R be a Banach ring. The kernel map $\mathcal{M}(R) \rightarrow \text{Spec}(R), x \mapsto \wp_x$ is continuous with respect to the Zariski topology on $\text{Spec}(R)$.

Proof. Let $D(f) = \{f \neq 0\} \subset \text{Spec}(R)$ be a principal open subset for some $f \in R$. The preimage of $D(f)$ under the kernel map is just the set $\{x \in \mathcal{M}(R) : |f|_x > 0\} = f^{-1}(\mathbb{R}_{>0})$, which is open in $\mathcal{M}(R)$ by definition of the topology on $\mathcal{M}(R)$. □

Example 1.19. Let us consider the spectrum $\mathcal{M}(\mathbb{Z})$ in Example 1.3. Under the kernel map $\mathcal{M}(\mathbb{Z}) \rightarrow \text{Spec}(\mathbb{Z})$, the points $|\cdot|_p^\infty$ for each prime number p are mapped to the prime ideal (p) , the other above points are all mapped to the zero ideal (0) .

Yang: Is this surjective? what is its fiber?

Yang: Seems i really need the surjectivity to analysis spectrum of $\mathbf{k}\{T\}$.

Proposition 1.20. Yang: To be added.

Construction 1.21. Suppose that R is a non-archimedean Banach ring with valuation subring R° and maximal ideal $R^{\circ\circ}$. For each $x \in \mathcal{M}(R)$, there is an induced homomorphism $R^\circ \rightarrow \mathcal{H}(x)^\circ$ between the valuation subrings. Furthermore, we have an induced homomorphism between the residue rings $\tilde{R} = R^\circ/R^{\circ\circ} \rightarrow \mathcal{K}_{\mathcal{H}(x)}$. This gives rise to the *reduction map*

$$\text{Red} : \mathcal{M}(R) \rightarrow \text{Spec}(\tilde{R}), \quad x \mapsto \ker(\tilde{R} \rightarrow \mathcal{K}_{\mathcal{H}(x)}).$$

Let $f : R \rightarrow S$ be a bounded homomorphism of non-archimedean Banach rings. **Yang:** The following diagram commutes:

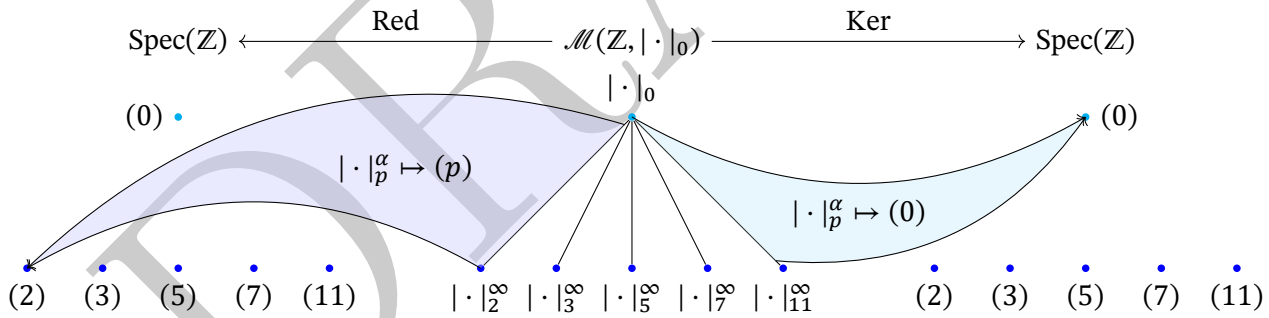
$$\begin{array}{ccc} \mathcal{M}(S) & \xrightarrow{\text{Red}} & \text{Spec}(\tilde{S}) \\ \mathcal{M}(f) \downarrow & & \downarrow \text{Spec}(\tilde{f}) \\ \mathcal{M}(R) & \xrightarrow{\text{Red}} & \text{Spec}(\tilde{R}) \end{array}$$

Hence the reduction map is functorial with respect to bounded homomorphisms of non-archimedean Banach rings.

Example 1.22. Let $(\mathbb{Z}, |\cdot|_0)$ be the Banach ring with the trivial norm. The reduction ring is $\tilde{\mathbb{Z}} = \mathbb{Z}$.

- $x = |\cdot|_p^\alpha$ for some prime number p and $\alpha \in (0, \infty]$: $\mathcal{K}_{\mathcal{H}(x)} \cong \mathbb{F}_p$ and the induced homomorphism $\tilde{\mathbb{Z}} = \mathbb{Z} \rightarrow \mathcal{K}_{\mathcal{H}(x)} = \mathbb{F}_p$ is the natural projection $\mathbb{Z} \rightarrow \mathbb{F}_p$;
- $x = |\cdot|_0$: $\mathcal{K}_{\mathcal{H}(x)} \cong \mathbb{Q}$ and the induced homomorphism $\tilde{\mathbb{Z}} = \mathbb{Z} \rightarrow \mathcal{K}_{\mathcal{H}(x)} = \mathbb{Q}$ is the natural inclusion $\mathbb{Z} \rightarrow \mathbb{Q}$.

The following diagram illustrates the reduction map and the kernel map for the spectrum $\mathcal{M}(\mathbb{Z}, |\cdot|_0)$:



Proposition 1.23. Let R be a non-archimedean Banach ring and $\tilde{U} \subset \text{Spec}(\tilde{R})$ be a Zariski open subset. Then the preimage $\text{Red}^{-1}(\tilde{U})$ is a closed subset of $\mathcal{M}(R)$.

Proof. **Yang:** To be completed. □

Proposition 1.24. Let $\mathbf{k}$ be a complete non-archimedean field and A, B a strictly $\mathbf{k}$ -affinoid algebras. Suppose that $\varphi : A \rightarrow B$ is a bounded $\mathbf{k}$ -algebra homomorphism. Then we have:

- if φ is finite, then the induced homomorphism $\tilde{\varphi} : \tilde{A} \rightarrow \tilde{B}$ is finite;
- if φ is of finite type, then the induced homomorphism $\tilde{\varphi} : \tilde{A} \rightarrow \tilde{B}$ is of finite type.

Yang: To be revised.

1.3 Spectrum of Tate algebras

In this subsection, fix an algebraically closed complete non-archimedean field $\mathbb{k}$.

Spectrum of Tate algebra in one variable Let $A = \mathbb{k}\{T/r\}$. We list some types of points in the spectrum $\mathcal{M}(A)$.

Construction 1.25. For each $a \in \mathbb{k}$ with $|a| \leq r$, we have the *type I* point x_a corresponding to the evaluation at a , i.e., $|f|_{x_a} := |f(a)|$ for each $f \in A$.

For each closed disk $E = E(a, s) := \{b \in \mathbb{k} : |b - a| \leq s\}$ with center $a \in \mathbb{k}$ and radius $s \leq r$, we have the point $x_E = x_{a,s}$ corresponding to the multiplicative semi-norm defined by

$$|f|_{x_E} = |f|_{x_{a,s}} := \sup_{b \in E(a,s)} |f(b)|$$

for each $f \in A$. If $s \in |\mathbb{k}^\times|$, then the point x_E is called a *type II* point; otherwise, it is called a *type III* point.

Let $E_n = E(a_n, s_n)$ be a sequence of closed disks in $\mathbb{k}$ such that $E_{n+1} \subsetneq E_n$ and $\bigcap_n E_n = \emptyset$. Then we have the point $x_{\{E_n\}} = x_{\{a_n, s_n\}}$ corresponding to the multiplicative semi-norm defined by

$$|f|_{x_{\{E_n\}}} = |f|_{x_{\{a_n, s_n\}}} := \inf_n |f|_{x_{E_n}}$$

for each $f \in A$. Such a point is called a *type IV* point.

Proposition 1.26. The points in the spectrum $\mathcal{M}(\mathbb{k}\{r^{-1}T\})$ can be classified into four types as described in [Construction 1.25](#).

Proof. Fix $x \in \mathcal{M}(\mathbb{k}\{r^{-1}T\})$, set

$$s = \inf_{a \in \mathbb{k}} |T - a|_x \leq r, \quad E = \{a \in \mathbb{k} : |T - a|_x = s\} \subset E(0, r).$$

Case 1. $E \neq \emptyset$ and $s = 0$.

By assumption, there exists $a \in E$ such that $|T - a|_x = 0$. Note that if $f(a) = 0$, then $T - a \mid f$ in $\mathbb{k}\{r^{-1}T\}$ and hence $|f|_x = |T - a|_x |g|_x = 0$. Then we have

$$|f(a)| = ||f(a)| - |f(a) - f|_x| \leq |f|_x \leq |f(a)| + |f(a) - f|_x = |f(a)|$$

for each $f \in \mathbb{k}\{r^{-1}T\}$, which implies that $|f|_x = |f(a)|$. Thus x is a type I point x_a .

Case 2. $E \neq \emptyset$ and $s > 0$.

Let $a \in E$. Note that for every $b \in \mathbb{k}$, we have

$$|a - b| \leq \max\{|T - a|_x, |T - b|_x\} = |T - b|_x.$$

First we show that $E = E(a, s)$. For each $b \in E(a, s)$, we have $|T - b|_x \leq \max\{|T - a|_x, |a - b|\} = s$, which implies that $b \in E$. Conversely, for each $b \in E$, we have $|a - b| \leq \max\{|T - a|_x, |T - b|_x\} = s$.

Let $f \in \mathbb{k}[T]$ be a polynomial. Write $f = \prod_{i=1}^n (T - c_i)$ for some $c_1, \dots, c_n \in \mathbb{k}$. Then we have

$$|f|_x = \prod_{i=1}^n |T - c_i|_x \geq \prod_{i=1}^n |b - c_i| = |f(b)|, \quad \forall b \in E.$$

I claim that for every $\varepsilon \in (0, 1)$, there exists $b \in E$ such that $\varepsilon|T - c_i|_x < |b - c_i|$ for each $i = 1, \dots, n$. Indeed, if $c_i \notin E$, then $|T - c_i|_x = |b - c_i|$ for each $b \in E$. Hence we only need to consider the case when $c_i \in E$. Since $\mathbb{k}$ is algebraically closed, $E(a, s) \setminus \bigcup_{i=1}^n E(c_i, \varepsilon s) \neq \emptyset$. Choose b in the set. Then we have

$$|f(b)| \geq \prod_{i=1}^n \varepsilon |T - c_i|_x = \varepsilon^n |f|_x.$$

Thus $|f|_x = \sup_{b \in E} |f(b)|$ for each polynomial $f \in \mathbb{k}[T]$. Since polynomials are dense in $\mathbb{k}\{r^{-1}T\}$, we have $|f|_x = \sup_{b \in E} |f(b)|$ for each $f \in \mathbb{k}\{r^{-1}T\}$. Therefore, x is the point $x_E = x_{a,s}$, which is of type II or type III depending on whether $s \in |\mathbb{k}^\times|$ or not.

Case 3. $E = \emptyset$.

Set $E_n = \{a \in \mathbb{k} : |T - a|_x \leq s + 1/n\}$ and $a_n \in E_n$ for each $n \in \mathbb{N}$. By the similar argument as in Case 2, we have $E_n = E(a_n, s + 1/n)$. Note that E_n is a decreasing sequence of closed disks with $\bigcap_n E_n = E = \emptyset$.

For $c \in \mathbb{k}$, there exists N such that $\forall n \geq N$, we have

$$c \notin E_n \implies |T - c|_x > |T - a_n|_x \implies |T - c|_x = |a_n - c|.$$

Thus

$$\inf_n |T - c|_{E_n} = \inf_n |a_n - c| = |T - c|_x.$$

By multiplicativity, we have $\inf_n |f|_{E_n} = |f|_x$ for each polynomial $f \in \mathbb{k}[T]$. And then by density of polynomials, the equality holds for each $f \in \mathbb{k}\{r^{-1}T\}$. Therefore, $x = x_{\{E_n\}} = x_{\{a_n, s_n\}}$ is of type IV. $\square$

Proposition 1.27. The completed residue fields of the four types of points in the spectrum $\mathcal{M}(\mathbb{k}\{r^{-1}T\})$ are described as follows:

- type I point x_a : $\mathcal{H}(x_a)$ is isomorphic to $\mathbb{k}$;
- type II point $x_{a,s}$: $\mathcal{H}(x_{a,s}) \cong \mathcal{K}_{\mathbb{k}}((t))$;
- type III point $x_{a,s}$: $\mathcal{K}_{\mathcal{H}(x_{a,s})} \cong \mathcal{K}_{\mathbb{k}}$ and the value group $|\mathcal{H}(x_{a,s})^\times|$ is generated by $|\mathbb{k}^\times|$ and s ;
- type IV point $x_{\{a_n, s_n\}}$: $\mathcal{H}(x_{\{a_n, s_n\}})$ is an immediate extension of $\mathbb{k}$.

Yang: To be checked.

Proof. Yang: To be completed. $\square$

Example 1.28. The completed residue field $\mathcal{H}(x_a)$ for a type I point x_a with $a \in \mathbb{k}$ and $|a| \leq r$ is isomorphic to $\mathbb{k}$. Yang: To be complete.

Example 1.29. Here we describe the reduction map and the kernel map for the spectrum $\mathcal{M}(A)$. The reduction ring is

$$\tilde{A} = \begin{cases} \mathcal{K}_{\mathbb{k}}[T], & \text{if } r \in |\mathbb{k}^\times|; \\ \mathcal{K}_{\mathbb{k}}, & \text{if } r \notin |\mathbb{k}^\times|. \end{cases}$$

If $r \notin |\mathbb{k}^\times|$, then the reduction map $\text{Red} : \mathcal{M}(A) \rightarrow \text{Spec}(\tilde{A}) = \text{Spec}(\mathcal{K}_{\mathbb{k}})$ is trivial, i.e., every point in $\mathcal{M}(A)$ is mapped to the zero ideal of $\mathcal{K}_{\mathbb{k}}$. Hence we only consider the case when $r \in |\mathbb{k}^\times|$. Without

loss of generality, we may assume that $r = 1$.

For each type I point x_a with $a \in \mathbb{k}$ and $|a| \leq 1$, the induced homomorphism $\tilde{A} = \hat{\mathbb{k}}[T] \rightarrow \hat{\mathbb{k}}_{\mathcal{H}(x_a)} = \hat{\mathbb{k}}_{\mathbb{k}}$ is given by $T \mapsto a \bmod \mathbb{k}^\circ$. Thus the reduction map $\text{Red} : \mathcal{M}(A) \rightarrow \text{Spec}(\tilde{A}) = \text{Spec}(\hat{\mathbb{k}}[T])$ sends the type I point x_a to the maximal ideal $(T - \tilde{a})$. For a type II or type III point $x_{a,s}$ with $a \in \mathbb{k}$ and $s \leq 1$, the homomorphism $A^\circ \rightarrow \mathcal{H}(x_{a,s})^\circ$

Yang: To be continued.

Example 1.30. Here we describe the kernel map for the spectrum $\mathcal{M}(A)$. If $x = x_a$ is a type I point corresponding to $a \in \mathbb{k}$ with $|a| \leq r$, then $\wp_{x_a} = (T - a)$. For type II, type III or type IV points x , note that if $f \in A$ is invertible, then $|f|_x > 0$ for each $x \in \mathcal{M}(A)$ since $|f|_x |f^{-1}|_x = |1| = 1$. By Yang: ref, we only need to consider elements in A of the form $T - c$ for some $c \in \mathbb{k}$ with $|c| \leq r$. Since $|T - c|_x \neq 0$ for all $c \in \mathbb{k}$ with $|c| \leq r$, we have $\wp_x = (0)$. To summarize, the kernel map $\mathcal{M}(A) \rightarrow \text{Spec}(A)$ is given by

$$\wp_x = \begin{cases} (T - a), & \text{if } x = x_a \text{ is a type I point;} \\ (0), & \text{if } x \text{ is a type II, type III or type IV point.} \end{cases}$$

Spectrum of Tate algebra in several variables Let $A = \mathbb{k}\{T\}$. To consider the spectrum $\mathcal{M}(A)$, inspired by Example 1.30, we first look at the image of the points in $\mathcal{M}(A)$ under the kernel map $\mathcal{M}(A) \rightarrow \text{Spec}(A)$.

For every $\mathfrak{p} \in \text{Spec}(A)$, since A is noetherian, $\mathfrak{p}$ is finitely generated, say $\mathfrak{p} = (f_1, \dots, f_n)$. On $E(0, 1) \subset \mathbb{k}^n$, $\mathfrak{p}$ defines a closed analytic subspace $V = V(f_1, \dots, f_n) = \{x \in E(0, 1) : f_i(x) = 0, i = 1, \dots, n\}$. By Yang: ref, V is nonempty. On $E(0, 1)$, we have the supremum norm $\|\cdot\|_{\text{sup}}$. It inherits the metric from $E(0, 1)$ and hence is a complete metric space.

According to the proof of Proposition 1.26, we should use prime divisor on $\text{Spec}(A/\mathfrak{p})$ to replace the role of points in one variable case.

Yang: To be continued.

2 Affinoid spaces

Fix a non-archimedean completed valued field $(\mathbf{k}, |\cdot|)$ throughout this section. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. It should be viewed as the non-archimedean analogue of an affine variety in algebraic geometry or an affine scheme in scheme theory. The goal of this section is to define a suitable Grothendieck topology on X such that we can define a structure sheaf of analytic functions on X .

2.1 Affinoid domains

Consider $X = \mathcal{M}(A)$ with $A = \mathbf{k}\{T_1, \dots, T_n\}$. Yang: Not every open subset of X gives an affinoid space, that is, the completion of the ring of analytic functions on that open subset is not necessarily an affinoid algebra. Yang: Right? example?

Definition 2.1. A closed subset $V \subseteq X$ is called an *affinoid domain* if there exists a $\mathbf{k}$ -affinoid algebra A_V and a morphism of $\mathbf{k}$ -affinoid algebras $\theta : A \rightarrow A_V$ satisfying the following universal property: for every bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\psi : A \rightarrow B$ such that the induced map on spectra $\mathcal{M}(\psi) : \mathcal{M}(B) \rightarrow X$ has its image contained in V , there exists a unique bounded homomorphism $\varphi : A_V \rightarrow B$ such that the following right diagram commutes:

$$\begin{array}{ccc} & V & \\ \swarrow & & \searrow \\ X & \xleftarrow{\mathcal{M}(\psi)} & \mathcal{M}(B) \end{array} \quad \rightsquigarrow \quad \begin{array}{ccc} & A_V & \\ \theta \nearrow & & \searrow \varphi \\ A & \xrightarrow{\psi} & B \end{array}$$

In this case, we say that V is represented by the affinoid algebra A_V .

Slogan A closed subset $V \subset X$ is an affinoid domain if the functor “ $\text{Mor}(-, V)$ ” is representable by a $\mathbf{k}$ -affinoid algebra A_V .

Yang: Why we consider closed subset rather than open subset?

Definition 2.2. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. An affinoid domain $V \subseteq X$ is called a *strict affinoid domain* if the representing $\mathbf{k}$ -affinoid algebra A_V is a strict $\mathbf{k}$ -affinoid algebra. Yang: To be revised.

Construction 2.3. Let $f = (f_1, \dots, f_n)$ be a tuple of elements in A and $r = (r_1, \dots, r_n)$ be a tuple of positive real numbers. Consider the closed subset of X :

$$X(\underline{f/r}) := \{x \in X : |f_i(x)| \leq r_i, 1 \leq i \leq n\}.$$

Such a closed subset is called a *Weierstrass domain* of X . Moreover, we can define a $\mathbf{k}$ -affinoid algebra

$$A\{\underline{f/r}\} := A\{f_1/r_1, \dots, f_n/r_n\}.$$

Yang: The domain $X(\underline{f/r})$ is represented by $A\{\underline{f/r}\}$.

Construction 2.4. Let $f = (f_1, \dots, f_n), g = (g_1, \dots, g_m)$ be two tuples of elements in A and $r = (r_1, \dots, r_n), s = (s_1, \dots, s_m)$ be two tuples of positive real numbers. Consider the following closed subset of X :

$$X(\underline{f/r}; \underline{g/s}^{-1}) := \{x \in X : |f_i(x)| \leq r_i, |g_j(x)| \geq s_j, 1 \leq i \leq n, 1 \leq j \leq m\}.$$

Such a closed subset is called a *Laurent domain* of X . Moreover, we can define a $\mathbf{k}$ -affinoid algebra

$$A\{\underline{f/r}; \underline{g/s}^{-1}\} := A\{f_1/r_1, \dots, f_n/r_n, g_1^{-1}/s_1, \dots, g_m^{-1}/s_m\}.$$

Yang: The domain $X(\underline{f/r}; \underline{g/s}^{-1})$ is represented by $A\{\underline{f/r}; \underline{g/s}^{-1}\}$.

Construction 2.5. Let $f = (f_1, \dots, f_n), g$ be elements in A such that the ideal generated by them is the whole algebra A . Set $p = (p_1, \dots, p_n)$ be a tuple of positive real numbers. We define the following

closed subset of X :

$$X(\underline{f/p}, g) := \{x \in X : |f_i(x)| \leq p_i |g(x)|, 1 \leq i \leq n\}.$$

Such a closed subset is called a *rational domain* of X . Moreover, we can define a $\mathbf{k}$ -affinoid algebra

$$A\langle \underline{f/p}, g^{-1} \rangle := A\left\langle \frac{f_1}{p_1 g}, \dots, \frac{f_n}{p_n g} \right\rangle,$$

which is the quotient of the Tate algebra

$$A\langle T_1, \dots, T_n \rangle$$

by the ideal generated by the elements $p_i g T_i - f_i$ for $1 \leq i \leq n$. There is a natural bounded homomorphism $\varphi : A \rightarrow A\langle \underline{f/p}, g^{-1} \rangle$ induced by the inclusion. It can be shown that the closed subset $X(\underline{f/p}, g)$ is an affinoid domain represented by the affinoid algebra $A\langle \underline{f/p}, g^{-1} \rangle$. **Yang: To be checked**

Yang: We have a sequence of inclusion:

$$\{\text{Weierstrass domains}\} \subseteq \{\text{Laurent domains}\} \subseteq \{\text{Rational domains}\} \subseteq \{\text{Affinoid domains}\}.$$

Proposition 2.6. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the natural bounded homomorphism $\varphi : A \rightarrow A_V$ is flat.

We have $\mathcal{M}(A_V) \cong V$.

Proposition 2.7. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space.

- (a) Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V , and let $\psi : A \rightarrow B$ be a bounded homomorphism of $\mathbf{k}$ -affinoid algebras. Set $Y = \mathcal{M}(B)$, and let $\mathcal{M}(\psi) : Y \rightarrow X$ be the induced map on spectra. Then the inverse image $\mathcal{M}(\psi)^{-1}(V) \subseteq Y$ is an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra

$$B \hat{\otimes}_A A_V.$$

- (b) Let $V_1, V_2 \subseteq X$ be two affinoid domains represented by the $\mathbf{k}$ -affinoid algebras A_{V_1}, A_{V_2} respectively. Then the intersection $V_1 \cap V_2 \subseteq X$ is an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra

$$A_{V_1} \hat{\otimes}_A A_{V_2}.$$

Yang: To be revised.

Theorem 2.8 (Gerritzen-Grauert theorem). Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Let $V \subseteq X$ be an affinoid domain. Then there exist finitely many rational domains $V_1, \dots, V_m \subseteq X$ such that $V = V_1 \cup \dots \cup V_m$. **Yang: To be checked.**

Definition 2.9. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . The *dimension* of V is defined as the Krull dimension of the affinoid algebra A_V :

$$\dim(V) := \dim_{\text{Krull}}(A_V).$$

Yang: To be revised.

2.2 The Grothendieck topology and structure sheaf

Theorem 2.10 (Acyclicity theorem). Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Let $\{V_i\}_{i \in I}$ be a finite covering of X by affinoid domains. For each finite subset $J \subseteq I$, set $V_J := \bigcap_{j \in J} V_j$, which is an affinoid domain by Proposition 2.7. Then the augmented Čech complex

$$0 \rightarrow A \rightarrow \prod_{i \in I} A_{V_i} \rightarrow \prod_{\substack{J \subseteq I \\ |J|=2}} A_{V_J} \rightarrow \prod_{\substack{J \subseteq I \\ |J|=3}} A_{V_J} \rightarrow \dots$$

is exact. Yang: To be checked.

Definition 2.11. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. The *site of affinoid domains* on X , denoted by X_{Aff} , is defined as follows:

- The underlying category has as objects the affinoid domains $V \subseteq X$, and the morphisms are given by the inclusion maps.
- A covering of an affinoid domain $V \subseteq X$ is a finite family of affinoid domains $\{V_i\}_{i \in I}$ such that $V = \bigcup_{i \in I} V_i$.
- The structure sheaf $\mathcal{O}_{X_{\text{Aff}}}$ is defined by setting, for each affinoid domain $V \subseteq X$ represented by the $\mathbf{k}$ -affinoid algebra A_V ,

$$\mathcal{O}_{X_{\text{Aff}}}(V) := A_V.$$

Yang: To be revised.

2.3 Relative boundary and interior

3 Analytic spaces